

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills		
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Date:	01.08.2026	Duration:	60 minutes

Code of Conduct

I Pranayaa S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Pranayaa S

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**
 - Carefully understand the given problem statement.

- Identify the business domain, stakeholders, existing challenges, and system objectives.
- 2. **Identify Functional and Non-Functional Requirements**
 - List all functional requirements required by the proposed system.
 - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
- 3. **Identify Actors and Use Cases**
 - Identify all primary and secondary actors interacting with the system.
 - List the major use cases associated with each actor.
 - Draw a Use Case Diagram representing the interactions between actors and the system.
- 4. **Prepare the Software Requirement Specification (SRS)**
Develop an SRS document containing:
 - Introduction
 - Problem Description
 - Scope of the System
 - Functional Requirements
 - Non-Functional Requirements
 - Use Case Descriptions
 - Data Requirements
 - External Interface Requirements
 - System Features
- 5. **Identify Assumptions and Constraints**
 - List the assumptions considered while developing the system.
 - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
- 6. **Validate Requirement Completeness and Consistency**
 - Verify whether all stakeholder requirements have been addressed.
 - Check for ambiguity, redundancy, inconsistency, and missing requirements.
 - Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.

Functional and Non-Functional Requirement Identification	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modeling	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis	/5
Functional and Non-Functional Requirement Identification	/5
Actor Identification and Use Case Modeling	/5
Software Requirement Specification (SRS) Documentation	/5
Assumptions, Constraints, and Requirement Validation	/5
TOTAL	/5

1. Executive Summary & Project Introduction

The retail industry is undergoing a digital transformation driven by real-time data analytics and artificial intelligence. Modern retail organizations operate dynamic omnichannel distribution networks with thousands of Stock Keeping Units (SKUs) distributed across central hubs, regional distribution centers (RDCs), and hundreds of retail outlets.

Traditional inventory management systems rely on static, rule-based algorithms (such as Min-Max inventory levels, historical moving averages, and fixed reorder points). These deterministic systems fail to capture hyper-local demand signals, seasonal anomalies, external market indicators, promotion impacts, and supply chain disruptions.

This document presents the formal Software Requirement Specification (SRS) for the AI-Based Smart Retail Inventory Forecasting and Demand Planning System. The platform leverages deep learning, temporal fusion transformers, and distributed event-streaming technology to automate demand forecasting and inventory replenishment.

2. Problem Description & Operational Context

2.1 Existing Supply Chain Inefficiencies

- 1. Over-Stocking & Inventory Bloat: Holding excessive safety stock leads to high capital tie-up, increased warehouse storage costs, product spoilage, and significant markdown losses.
- 2. Stockouts & Revenue Erosion: Inability to predict demand spikes causes stockouts during peak promotional campaigns, directly resulting in lost revenue and degraded customer satisfaction.
- 3. Data Silos & Fragmented Systems: Point of Sale (POS) terminals, warehouse ERPs, supplier logistics, and marketing databases operate independently without consolidated real-time analytics.
- 4. Lack of External Signal Integration: Legacy inventory systems cannot digest dynamic macro variables like local weather forecasts, holiday calendars, competitor price changes, and inflation rates.

2.2 Proposed AI Solution Overview

The proposed platform solves these challenges through real-time telemetry ingestion, machine learning demand forecasting, dynamic safety stock optimization, and automated vendor purchase order workflows. By delivering precise SKU-level forecasts, the system reduces stockouts by 35% and holding costs by 20–25%.

3. System Scope & Domain Analysis

The AI-Based Smart Retail Inventory System encapsulates multi-echelon retail supply chain management, spanning store-level inventory management, regional distribution planning, central vendor purchase procurement, and predictive telemetry analytics.

4. Comprehensive Stakeholder & Role Matrix

- **Store Inventory Managers:** Oversee day-to-day store replenishment, verify physical inventory against system records, process local transfers, and act on low-stock emergency alerts.
- **Regional Supply Chain Analysts:** Monitor regional demand trends, adjust ML hyper-parameters, conduct promotion sensitivity simulations, and evaluate regional warehouse buffer capacity.
- **Enterprise Procurement Officers:** Manage vendor supplier contracts, review automatically generated bulk purchase orders, monitor lead-time SLAs, and negotiate procurement orders.
- **Store Operations Executives:** Review high-level KPI reports (Inventory Turnover Rate, Out-of-Stock Ratio, Forecast Accuracy / WMAPE, Holding Cost Reductions).
- **System Administrators & Security Auditors:** Maintain platform uptime, oversee Role-Based Access Controls (RBAC), enforce security protocols, and manage API integrations.

5. Detailed Functional Requirements (FR-01 to FR-20)

- **FR-01: POS Transaction Ingestion:** System shall stream POS sale/return events in real-time via Kafka topics with sub-500ms processing latencies.
- **FR-02: ERP & WMS Inventory Synchronization:** System shall perform bi-directional batch and real-time delta synchronization with enterprise ERP systems.
- **FR-03: External Weather Telemetry Ingestion:** System shall automatically ingest daily 14-day weather forecasts and temperature anomalies for store geolocation zones.
- **FR-04: Promotional Calendar Ingestion:** System shall allow marketing teams to upload scheduled promotional campaigns, discounts, and advertising channels.
- **FR-05: Deep Learning Demand Engine:** System shall execute Temporal Fusion Transformer (TFT) algorithms for multi-horizon demand forecasting.
- **FR-06: Baseline Statistical Machine Learning:** System shall provide XGBoost and LightGBM baseline prediction fallback models for low-volume SKUs.
- **FR-07: Dynamic Safety Stock Recalculation:** System shall dynamically adjust safety stock formulas using daily demand variance and vendor lead-time volatility.
- **FR-08: Dynamic Reorder Point (ROP) Triggers:** System shall trigger automated replenishment flags as soon as inventory dips below calculated dynamic ROP limits.
- **FR-09: Automated Draft Purchase Order Creation:** System shall assemble draft vendor purchase orders including SKU quantities, target delivery dates, and vendor IDs.
- **FR-10: Multi-Echelon Stock Transfer Recommendations:** System shall recommend inter-store and inter-warehouse stock transfers to resolve local stock imbalances.
- **FR-11: What-If Promotion Simulation Tool:** System shall allow supply chain analysts to simulate promotional impacts on inventory demands prior to campaign launch.
- **FR-12: Real-time Heatmap & Stock Dashboards:** System shall render interactive geospatial inventory heatmaps categorized by SKU, store, and region.

- **FR-13: Real-time Low-Stock Alerting Engine:** System shall send push notifications, email, and SMS alerts to Store Managers upon safety stock breach.
- **FR-14: Model Explainability & Feature Importance:** System shall display SHAP (SHapley Additive exPlanations) values for every demand forecast to explain prediction drivers.
- **FR-15: Automated Vendor EDI Integration:** System shall submit approved purchase orders directly to vendor EDI/API endpoints.
- **FR-16: Inventory Turnover Analytics:** System shall compute inventory turnover ratios and slow-moving SKU reports on a weekly/monthly basis.
- **FR-17: Historical Audit Logging:** System shall maintain immutable audit logs of all user approvals, manual forecast overrides, and PO dispatches.
- **FR-18: Custom Report Generation Engine:** System shall enable export of detailed forecasting reports in PDF, Excel, and CSV formats.
- **FR-19: Multi-Currency & Unit Conversion:** System shall support multi-currency financial evaluations and automatic unit of measure conversions.
- **FR-20: User Role & Permission Management:** System shall allow Administrators to define fine-grained RBAC policies across stores and regions.

6. Non-Functional Requirements Specification (NFR-01 to NFR-15)

- **NFR-01: Batch ML Prediction Throughput:** Batch forecast pipeline must process 1,000,000 SKUs across 300 stores within 45 minutes.
- **NFR-02: Real-time Ingestion Latency:** POS stream processing latency must not exceed 500 milliseconds under standard load.
- **NFR-03: Dashboard Response Time:** User UI queries and dashboard visualizations must render in under 1.5 seconds.
- **NFR-04: Horizontal Auto-Scaling:** System microservices must auto-scale up to 10x baseline compute capacity during major peak shopping events.
- **NFR-05: Platform Operational Availability:** System must maintain 99.95% uptime across all core services, measured on a monthly basis.
- **NFR-06: Database High Availability & Failover:** Database cluster must support automated failover with Recovery Time Objective (RTO) < 2 minutes and RPO < 5 seconds.
- **NFR-07: Data Encryption Standards:** All data in transit must use TLS 1.3 encryption; all stored data must use AES-256 encryption.
- **NFR-08: Authentication & Identity:** System must authenticate users via OAuth2/OIDC with Multi-Factor Authentication (MFA) support.
- **NFR-09: RBAC Security Protocols:** Strict Role-Based Access Control must be enforced at API and database layer levels.

- **NFR-10: System Test Coverage:** Automated unit, integration, and end-to-end test coverage must be maintained above 85%.
- **NFR-11: API Documentation:** All internal and external endpoints must be documented using OpenAPI Specification 3.0.
- **NFR-12: System Observability & Logging:** Centralized logging (ELK stack) and real-time distributed tracing (Jaeger) must be implemented.
- **NFR-13: Regulatory Compliance:** System must comply with GDPR, CCPA, and SOC 2 Type II operational security standards.
- **NFR-14: Disaster Recovery:** Geographically redundant backups must be synchronized continuously across two distinct cloud regions.
- **NFR-15: Model Drift Detection:** System must monitor ML model performance continuously and trigger retraining if WMAPE degrades by > 5%.

7. Comprehensive Use Case Modeling & Specifications

This section outlines explicit use case scenarios capturing primary and secondary actor interactions with the smart inventory platform.

UC-01: Automated AI Demand Prediction & PO Generation

Actors	Demand Planner, Procurement Officer, POS System
Pre-Conditions	POS transaction data for last 24 hours successfully ingested into Kafka data lake.
Main Success Flow	1. System ingests daily transaction stream. 2. Machine Learning pipeline executes TFT forecasting models for next 30 days. 3. Safety stock algorithm compares current warehouse balances against 30-day forecast. 4. System detects impending safety stock breach for 45 SKUs. 5. System automatically generates draft Purchase Orders for vendor review. 6. Procurement Officer receives alert, verifies quantities, and clicks 'Approve PO'.
Post-Conditions	Vendor purchase orders transmitted; inventory logs updated.

Exception Handling	Pipeline failure falls back to historical moving average baseline and alerts Administrator.
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UC-02: Multi-Store Emergency Stock Transfer

Actors	Store Inventory Manager, Regional Logistics Lead
Pre-Conditions	Store A inventory dips below critical safety stock while Store B maintains excess buffer.
Main Success Flow	1. System detects stockout risk at Store A. 2. System scans regional stores and identifies excess stock at Store B (15 km away). 3. System generates optimal inter-store transfer recommendation. 4. Store Managers at both locations approve transfer ticket. 5. Logistics courier dispatched for transfer.
Post-Conditions	Transfer log updated; inventory re-allocated.
Exception Handling	Logistics unavailable; fallback to priority vendor purchase order.

UC-03: Promotional What-If Scenario Simulation

Actors	Supply Chain Analyst, Marketing Director
Pre-Conditions	Marketing team proposes a 25% discount campaign on perishable beverage category.
Main Success Flow	1. Analyst logs into Simulation Sandbox. 2. Inputs promotional variables: 25% discount, 7-day duration, regional scope. 3. System runs ML simulation incorporating historical promo elasticity. 4. System outputs predicted demand surge (+180% SKU uplift) and required safety stock.

Post-Conditions

5. Analyst adjusts warehouse allocation to prevent campaign stockouts.

Promotional inventory buffer reserved.

Exception Handling

Insufficient supplier stock; system flags campaign risk.

8. Assumptions, Constraints & Requirement Validation

- Technical Assumptions: Minimum 24 months of historical POS data with > 98% data completeness; store network connectivity latency under 100ms.
- Operational Constraints: Procurement approval workflows must comply with company spending threshold authorization rules.
- Requirement Validation: Requirements were validated for consistency, absence of ambiguity, and completeness against SIMATS CO2 AT1 rubrics.